MILITARY SPECIFICATION

OSCILLOSCOPE, ENGINE IGNITION ANALYZING, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable, engine ignition analyzing oscilloscope.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more oscilloscopes for first article inspection (see 4.3). The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The oscilloscope shall be designed specifically for automotive engine diagnosis, tune-up, and service. The oscilloscope shall be solid state and shall be capable of testing two and four cycle engines with 12 and 24 volt negative ground electrical systems. The oscilloscope shall be capable of testing both conventional distributor breaker point and solid-state, breakerless, high energy ignition systems, including capacitive discharge types. The oscilloscope shall be portable and shall operate from a power source of 115 volts, 60 hertz (Hz), single phase (see 3.5.2).

3.2.1 Hazard prevention. Oscilloscope circuits shall be either electrically isolated or shall include protective features so that a failure in the oscilloscope, its harness, or cables shall not damage the ignition circuits of the vehicle under test. Provision shall be made for a grounding connection between the block of the engine being tested and the case and control panel of the oscilloscope (see 3.2.5).

3.2.2 Electrical protection. The oscilloscope shall be protected from damage in the event of internal failure resulting in high currents and from attempted operation on improper voltages. A fuse holder, if used, shall be located so that fuses can be replaced. The fuse holder shall not protrude beyond other components on the same surface of the oscilloscope. There shall be provision for inclusion of spare fuses (one of each type used) with the oscilloscope. If the protection is provided by other than fuses, the reset control shall be accessible and shall not protrude beyond other components on the same surface of the oscilloscope. The control shall be capable of being reset without tools of any kind and shall be clearly marked.

* 3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.5 Safety and health requirements. The oscilloscope shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the oscilloscope itself. The oscilloscope shall also comply with

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applicable requirements of MIL-STD-454 and NFPA 70. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 3.2.1, 6.2.1, and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.4 Construction. The oscilloscope shall be furnished complete so that it can be used for the operations specified herein. Parts used in fabricating the oscilloscope shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance. Non-functional sharp edges, points, and projections shall be eliminated or guarded.

* 3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.2 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

* 3.4.3 Mounting of components. The chassis and panel shall be one assembly on which all component parts are mounted. Parts shall be accessible for servicing. The chassis and panel assembly shall be constructed to allow positioning the oscilloscope on its bottom, top, and side during servicing without causing injurious mechanical stress or permanent deformation of the assembly or any component. Circuit boards shall be installed in a manner to prevent water or fluid from standing on them.

3.5 Performance and product characteristics.

* 3.5.1 Stability. The oscilloscope shall be capable of maintaining a stable pattern within an engine operating range of 500 to 3,000 crankshaft revolutions per minute (rpm) without manual control or adjustment. After a ten minute warm-up, the cathode ray tube (crt) shall maintain a stable pattern at any engine speed between 500 to 3000 rpm without manual control or adjustment. Vertical and horizontal pattern movement (drift) shall not exceed $\pm$ five percent of the usable crt display grid height or width. For the purpose of this requirement, it is assumed that the synchronizing device signal and the pattern generating device (magneto or distributor) have no "jitter" incidental to engine gear backlash. Patterns on the oscilloscope shall be clear and essentially free from deflections due to stray voltages and fields.

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3.5.2 Power. The oscilloscope shall perform its required functions when operated on a power source of 110 to 125 volts, 60 Hz, single phase alternating current or, when specified (see 6.2.1), on 6, 12, or 24 volts direct current.

3.5.3 High voltage supply regulation. Regulation of the high voltage supply shall be such as to produce a spot size and intensity combination that satisfies the requirements of 3.5.7. The high voltage shall be as recommended by the tube manufacturer for visual observation of medium and fast repetitive signals.

* 3.5.4 Horizontal sweep synchronization. The horizontal sweep shall be synchronized by secondary pickup of the engine's number one spark plug as the usual method, or from any other spark plug when desired or required.

3.5.5 Constant horizontal sweep length circuit. A circuit shall be provided to prevent sweep length from varying as engine speed is changed. The circuit shall maintain the length of the horizontal sweep 720 degrees, or expanded (see 3.6.1.4), within ± five percent when the engine speed is varied from 500 to 3,000 rpm.

3.5.6 Accuracy. Except for the first and last ten percent of the trace, a linear relationship shall exist between the horizontal trace and the engine crankshaft position throughout an engine speed range of 500 to 3,000 rpm.

3.5.7 Wave forms. Vertical deflection, as produced by ignition events, shall be obtained by either primary or secondary ignition circuit pickup, or both, as required by the specific circuitry of the oscilloscope. The wave forms shall be clear, well defined, and stable. The wave forms shall permit determination of the total available voltage from the coil, the voltage required to overcome rotor and spark plug gaps, and the resistance in the engine secondary circuit. The wave forms shall present sufficient information to permit location and identification of the following ignition circuit events, with the engine operating under various load conditions throughout its respective speed range:

(a) Open or shorted portions of ignition circuits resulting from faulty insulation, flash-over, defective coils, defective capacitors, and defective ignition harness.
(b) Leaking, fouled, or shorted spark plugs.
(c) Arcing or bouncing breaker points.
(d) Defective cam, camshaft, rotor, bearings, or cam follower (magneto and distributor) affecting point opening, point closing, and dwell in distributor systems.
(e) Measurement of dwell angle.

3.5.8 Operating temperature. The oscilloscope and accessories shall be capable of withstanding without damage, and shall operate within the requirements set forth in 3.5.1 and 3.5.7, while being stabilized to an operating temperature of +30° to +35°F and +110° to +115°F.

3.5.9 Humidity conditions. The oscilloscope and accessories shall be capable of withstanding the storage specified below without damage and shall operate within the requirements of 3.5.1 and 3.5.7:

(a) After having been subjected to 20 hours exposure to, and while at, a relative humidity of 93 to 97 percent in an air temperature of +73° to +77°F, air movement and radiation negligible.

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(b) After having been subjected to four hours exposure to and while at a relative humidity of 5 to 10 percent in an air temperature of +110° to +115°F, air movement and radiation negligible.

3.5.10 Dielectric strength. The oscilloscope and accessories shall withstand without damage 1,000 volts root mean square (rms) at 60 Hz between mutually insulated circuits, and between insulated circuits and enclosure.

3.6 Components.

3.6.1 Operating controls. Operating controls shall be located on the front panel. The controls may be separate or integral with other operating controls (concentric shaft construction). Knob controls shall be mounted with their shafts perpendicular to the panel. The function of each control shall be permanently identified by words or standard abbreviations in colors that contrast with the panel color.

3.6.1.1 Power switch. The power switch shall have an OFF position clearly marked.

3.6.1.2 Indicator light. An indicator lamp assembly shall be provided to indicate when power is being applied to the oscilloscope. When lit, the indicator light shall be visible within a minimum viewing angle of 120 degrees from the front of the oscilloscope.

3.6.1.3 Vertical gain control (height). A control shall be provided to adjust the amplitude of the vertical deflections produced by the ignition signals. The controlled gain shall be high enough to amplify the vertical displacement sufficient to extend at least two-thirds on the crt face.

* 3.6.1.4 Pattern expand control (sweep). A control shall be provided to control the gain of the horizontal circuits so that sweep length may be adjusted to any point within the range delineated by the following:

(a) Presentation of a single trace across the crt, which portrays engine ignition events occuring during a complete engine cycle of 720 degrees (4-cycle engine).

(b) Presentation of a portion of the complete 720 degree trace horizontally expanded across the crt. The control shall allow the expanded portion to cover a crankshaft rotation of at least 130

degrees; each degree to be represented by a distance of 0.03125 inch along the base line. The point of trace initiation shall not move more than 0.50 inch when sweep length is changed from a single presentation to the expanded portion thereof, or vice versa, within an engine speed range of 500 to 3,000 rpm.

3.6.1.5 Horizontal centering control (parade). A control shall be provided for horizontal centering of the complete 720 degree trace (see 3.6.1.4(a)) on the face of the crt. When the trace is viewed in expanded portions (see 3.6.1.4(b)), the control shall provide for parading the entire expanded trace across the face of the crt to allow viewing ignition patterns related to the individual cylinders.

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3.6.1.6 Vertical centering control. A control shall be provided to adjust the vertical position of the crt sweep to any position between the top and bottom of the crt screen.

3.6.1.7 Focus control. A control shall be provided to focus the crt beam, not exceeding 0.0156 inch diameter, to meet the requirements of 3.5.7. This control may be either a service or an operating control.

3.6.1.8 Beam intensity control. A control shall be provided to adjust the intensity of the crt beam to meet the requirements of 3.5.7. This control may be either a service or an operating control.

3.6.2 Service controls. Service controls shall be clearly labeled. Those service controls located on the instrument panel shall be of the screwdriver adjustment type. If located elsewhere, the service controls shall be protected by a removable cover and located within the protected area of the opening. Service controls shall include controls for periodic adjustments necessary to obtain satisfactory performance from the oscilloscope, but which do not normally require adjustment during operation.

3.6.3 Presentation components.

* 3.6.3.1 Cathode ray tube. The oscilloscope crt shall have an actual usable display pattern area of at least 4.06 inches wide by three inches in height.

* 3.6.3.2 Light shield. A partial hood shall be attached to the enclosure at the top of the crt to improve contrast under bright light conditions.

3.6.4 Electron tubes. With the exception of the crt, solid state devices shall be used in place of electron tubes.

3.6.5 Rectifiers. Solid state rectifiers shall be used in all applications.

3.6.6 Internal cabling. Insulated wires shall be neatly formed and in accordance with standard commercial electrical practice. Internal wiring shall be color coded.

* 3.6.7 Connecting harness or leads. Unless otherwise specified (see 6.2.1), the connecting harness or leads for the oscilloscope shall be a minimum of 15 feet in length. The leads shall be protected from sharp bends for no less than six inches where the leads enter the cabinet.

3.6.8 Enclosure. The oscilloscope enclosure shall be provided with a carrying handle. Resilient, oil-resistant rest buttons shall be located at each corner of the base of the oscilloscope and at each corner of the side opposite the carrying handle.

3.6.8.1 Storage compartment. A compartment shall be provided for storage of harnesses, instruction handbook, leads, and adapters. This compartment may be a part of the enclosure or a separate individual unit. If this compartment is an integral part of the enclosure, it shall be completely accessible. If separate, the compartment shall be completely enclosed and provided with a handle. Resilient, oil-resistant rest buttons shall be located at each corner of the base.

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* 3.6.9 Power supply cord. Unless otherwise specified (see 6.2.1), the oscilloscope shall be furnished with not less than 15 feet of three conductor, flexible, electrical cord. The cord shall be No. 18 AWG, types SO, SJO, or STO, heavy duty, conforming to NFPA 70 and shall terminate in a three prong polarized connector plug, either the cord-grip type or molded-on type.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Optional equipment. Optional equipment and accessories, when required, shall be furnished as specified (see 6.2.1).

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The oscilloscope shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.11 Workmanship. Standards of workmanship shall assure that the oscilloscope shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Operational test (see 4.6.1).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	100 percent inspection	-
Operational test (4.6.1)	II	1.0
Dielectric strength test (4.6.4)	II	1.0
Packaging inspection (4.7)	S-4	4.0

* 4.5 Product Examination. Visually and manually examine the oscilloscope to determine conformance with the requirements of 3.2 through 3.3, 3.4 through 3.4.3, and 3.6.2 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts, such as controls and hookups, by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the oscilloscope and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

* 4.6.1 Operational test. This test shall be performed in an ambient air temperature of +70° to +90°F. An automotive engine, or an equivalent mockup, shall be used in performing this test. Checks shall be made over an engine speed range of 500 to 3,000 rpm. The oscilloscope shall be checked to determine compliance with 3.5.1 through 3.5.6. The oscilloscope shall correctly portray a representative pattern of each of the malfunctions specified in 3.5.7 as well as normal engine operation. Controls shall be checked to determine compliance with 3.6.1 through 3.6.1.8. Portrayals obtained shall be checked for accuracy by comparison with sample patterns shown in the manufacturer's technical data (see 6.3).

4.6.2 Operating temperature tests.

4.6.2.1 High operating temperature test. The oscilloscope and accessories, after being in an ambient air temperature of +110° to +115°F until stabilized, and while at that temperature, shall be tested in accordance with 4.6.1 (see 3.5.8).

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4.6.2.2 Low operating temperature test. The oscilloscope and accessories, after being in an ambient air temperature of +30° to +35°F until stabilized, and while at that temperature, shall be tested in accordance with 4.6.1 (see 3.5.8).

4.6.3 Overload test. The overload protective device shall be tested for its ability to operate in sufficient time to prevent damage to the oscilloscope when an overload of 25 percent or more is applied (see 3.2.2).

4.6.4 Dielectric strength test. The oscilloscope and accessories shall be subjected to 1,000 volts rms at 60 Hz between insulated circuits and ground, and between mutually insulated circuits, in accordance with method 301 of MIL-STD-202. The oscilloscope shall then be inspected for evidence of damage and tested in accordance with 4.6.1 (see 3.5.10).

4.6.5 Humidity test. The oscilloscope and accessories shall be subjected to 20 hours exposure with relative humidity between 93 and 97 percent in an air temperature of +73° to +77°F, air movement and radiation negligible. While at this condition, the oscilloscope shall be tested in accordance with 4.6.1 to determine compliance with 3.5.9. Following the above, the oscilloscope shall be subjected to four hours exposure at a relative humidity of 5 to 10 percent in an air temperature of +110° to +115°F, air movement and radiation negligible. While at this condition, the oscilloscope shall be tested in accordance with 4.6.1 to determine compliance with 3.5.9.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-E-17555.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The oscilloscope covered by this specification is used for detecting, identifying, and locating normal and abnormal engine ignition events. The oscilloscope, used with connecting harness, is suitable for checking sealed and suppressed, as well as conventional, single ignition systems employed on automotive engines, and on out-board engines of amphibious equipment such as boats and pontoon barges. Interpretation of wave patterns presented by the oscilloscope allows detection of impending failures or malfunction of ignition components. When used in conjunction with a proper transducer, the oscilloscope will present interpretative data which is directly convertible to proportionate electrical impulses.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.5 and 6.4).
(d) Direct current voltage requirements when required (see 3.5.2).
(e) Harness length (leads) if different (see 3.6.7).
(f) Power supply cord if different (see 3.6.9).
(g) Optional equipment when required (see 3.8).
(h) Painting and finishing if different (see 3.9).
(i) Product identification if different (see 3.10).
(j) First article inspection if different (see 4.3).
(k) Quality conformance inspection if different (see 4.4).
(l) Sampling when required (see 4.4.1).
(m) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 4.6.1 and 6.3.1).

* 6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply for Manufacturers (FSCM), nomenclature, part number and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - YD
Air Force - 99

Review activity:

Air Force - 8?

User activities:

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Preparing activity:

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-45801F	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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